CHAPTER 01

Introduction

What management science is, how it developed, and how an analysis actually proceeds

TEXTBOOK	CHAPTER	WEEK
Modern Management Science, 9th ed.	**Ch. 1, pp. 3–19**	**Week 1**

Management Science · Fall 2026 · Kitae Kim, School of Business, Korea Aerospace University

CHAPTER CONTENTS

Five sections, one question

The chapter is built around a single question: when a manager faces a decision, what does a scientific approach actually add?

1 **The development of quantitative decision making** — Where the field came from — from Taylor to operations research — p. 3

2 **Characteristics of management science** — The five features that distinguish it from other approaches — p. 5

3 **The steps of management science analysis** — Figure 1.3 — the analysis cycle, and why it loops — p. 9

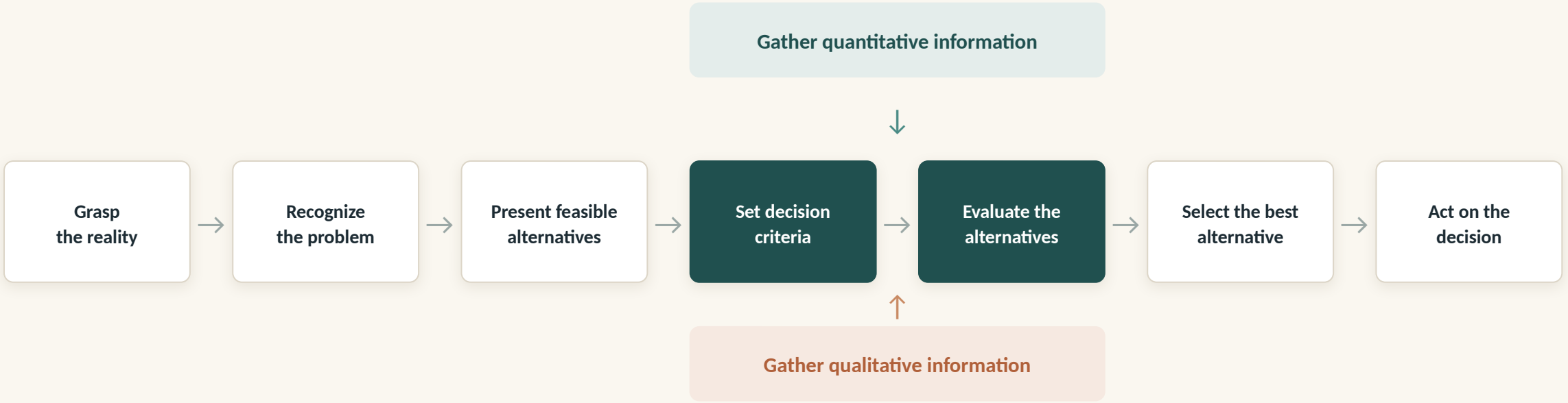
4 **Management science techniques** — The toolbox, and which tools this course covers — p. 11

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FIGURE 1.1

The systematic decision-making process

Every decision follows this shape, whether or not anyone writes it down. Management science makes the middle steps explicit.



Information is gathered only once the alternatives are on the table — and both kinds matter. Numbers alone do not set the criteria, and judgement alone does not evaluate the options. Management science works on the quantitative half without pretending the other half does not exist.

Five people who made decisions calculable

Before the field had a name, individual problems were being solved mathematically — each one still taught today.

1909

Erlang

Queueing theory

Telephone traffic: how many lines are enough? Waiting made calculable

1911

Taylor

Scientific management

Time-and-motion study — measuring work instead of estimating it

1913

Harris

The EOQ model

Economic order quantity — the first inventory optimization model

1914

Lanchester

Attrition models

Differential equations for the outcome of a military engagement

1917

Gantt

The Gantt chart

Scheduling made visible — still on every project plan today

What they share: each replaced a judgement call with a model that could be checked. None of them called it management science — the name comes later, and only after a war made the approach unavoidable.

A war gave the field its name and its method

DURING THE WAR

Operations Research

British and American teams were asked military questions that had no precedent: how should radar stations be sited, how large should a convoy be, where should depth charges be set to explode.

The answers came from interdisciplinary teams — physicists, mathematicians, economists — working on operational data rather than laboratory data. That working method is the one the field kept.

AFTER THE WAR

Management Science

The same teams turned to industry. Dantzig's simplex method (1947) made linear programming solvable at scale, and the arrival of the computer made large models practical.

In business the field took the name management science; in engineering and the military it stayed operations research. The two names describe the same discipline.

The simplex method is where this course begins in earnest: Chapter 2, next week. Everything in Weeks 2–7 rests on it.

SECTION 2

Five characteristics of management science

01

A focus on managerial problems

The subject is a decision someone actually has to make, not a phenomenon to be described. The test of the work is whether the decision improves.

02

A systems approach

The unit of analysis is the whole system, not the department. Optimizing one part in isolation routinely makes the whole worse.

03

The scientific research method

Observe, formulate, model, test against data, revise. A model that has not been validated is a hypothesis, not an answer.

04

Interdisciplinary teamwork

Real problems do not respect subject boundaries. The wartime teams mixed disciplines deliberately, and the field kept the habit.

05

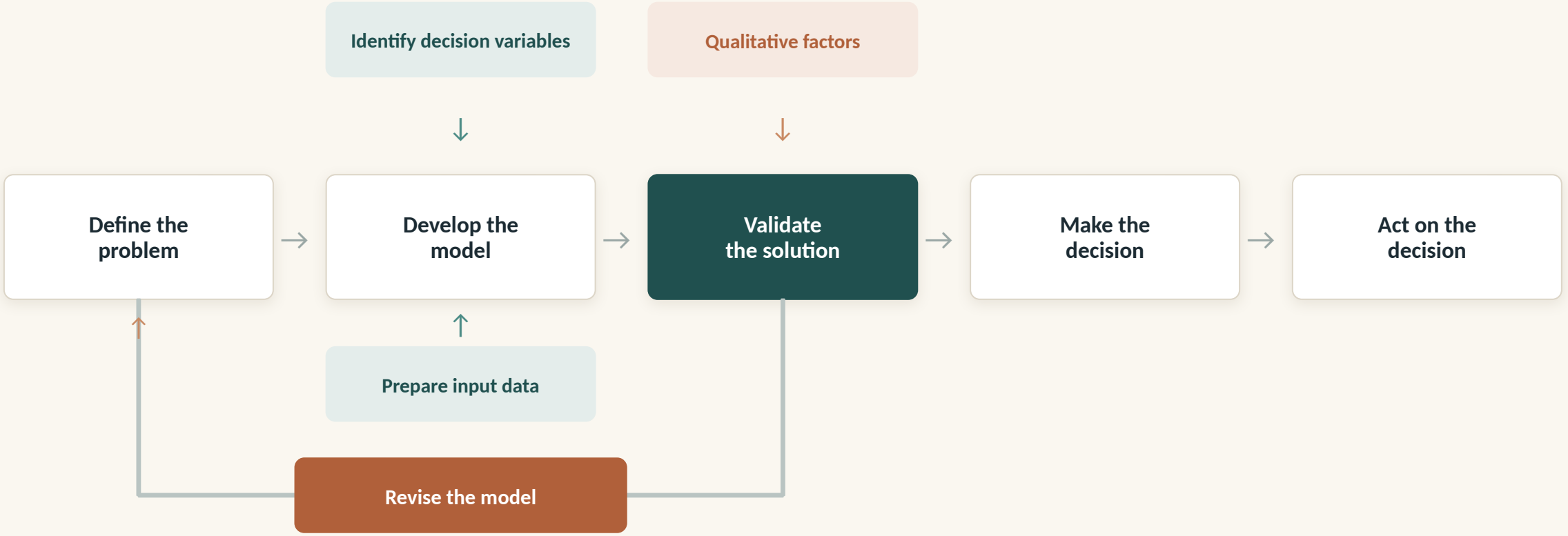
Mathematical methods and computers

Models large enough to be realistic are too large to solve by hand. This is why Week 4 puts you in Python.

Notice that only the last of the five is about mathematics. The other four are about how you frame the problem and who you frame it with.

FIGURE 1.3

The steps of a management science analysis



The loop is the point. A first model is almost never right — validation sends you back to the definition, not forward to the decision.

SECTION 4

The toolbox — and what this course covers

Linear programming	Ch. 2–5 · Weeks 2–7	
Network models and transportation	Ch. 6 · Week 10	
Integer linear programming	Ch. 7 · Week 9	
Multi-criteria decision making	Ch. 8 · Weeks 11–12	
Decision analysis	not covered this term	
Simulation	not covered this term	
Forecasting	not covered this term	
Markov process models	not covered this term	
Project scheduling: PERT / CPM	not covered this term	
Inventory systems	not covered this term	
Queueing systems	not covered this term	
Game theory	Ch. 11 · Week 13	

Eleven techniques are listed in the chapter; this course covers four of them plus game theory, which the chapter's list does not mention. The rest are the natural follow-on reading.

SECTION 5

Where these models actually live

A model that runs once in a classroom is an exercise. A model that runs every night against live data is a management science system.

01

Transaction systems

Sales, inventory, and operations data recorded as the business runs

→

02

Database

The organized record that analysis draws on

→

03

The model

Optimization, forecasting, or simulation applied to that data

→

04

Decision support

Results delivered to a manager who still makes the call

Why it matters for the model

The quality of a decision is bounded by the quality of the data behind it. Much of the effort in a real project goes into the third step of Figure 1.3 — preparing input data — not into the mathematics.

Why it matters for you

This is the reason the course is taught in Python on Google Colab rather than by hand. From Week 4 you will be building models that read data, not models that fit on a whiteboard.

APPLICATION CASE · P. 17

Taco Bell's SMART labor management system

3

models combined

forecasting, simulation, and integer programming

15 min

forecast interval

customer transactions predicted in quarter-hour buckets

1996

Edelman Award

INFORMS Franz Edelman Award for applied operations research

The problem

A fast-food restaurant's demand swings enormously within a single day, and labor is the largest controllable cost. Schedule too many crew and you burn margin; too few and service collapses at exactly the busiest moment.

The management science answer

Forecast transactions per 15-minute interval, simulate how the restaurant behaves under a given crew, then solve an integer program for the schedule that meets the service target at least cost. Three techniques from this chapter's list, working as one system.

Source: textbook Ch. 1 application case, p. 17; Hueter & Swart, “An Integrated Labor-Management System for Taco Bell,” Interfaces 28(1), 1998; INFORMS Franz Edelman Award. Verify reported figures before citing them.

CHAPTER 1 · IN SUMMARY

Four things to take away

01	**Management science is about decisions**	Not description, not explanation — the work is judged by whether the decision it supports is better.
02	**The method came from solving real problems**	From Taylor's stopwatch to wartime operations research to Dantzig's simplex method in 1947.
03	**The analysis is a loop, not a line**	Figure 1.3 returns from validation to problem definition. Expect to go round it more than once.
04	**Quantitative does not mean exclusive**	Figure 1.1 puts qualitative information in the diagram on purpose. The model informs the decision; it does not make it.

Next week · Chapter 2 — Linear optimization models and linear programming